

Mathematical Analysis. Lecture 19 不定积分.

1. 原函数与不定积分的概念.

定义: 假设在区间 I 上定义的两个函数 $F(x)$ 以及 $f(x)$ 满足 $F'(x) = f(x)$ 或 $dF(x) = f(x)dx$.

则 $F(x)$ 是 $f(x)$ 在区间 I 上的一个原函数. 可以有无穷多个原函数.

定义: $f(x)$ 在区间 I 上原函数全体称为 $f(x)$ 在 I 上的不定积分, 记作 $F(x) = \int f(x) dx$.

若 $F'(x) = f(x)$, 则 $\int f(x) dx = F(x) + C$. (C 为任意常数). **C 不能丢!**

e.g. 求 $\int e^x dx = e^x + C, C \in \mathbb{R}$.

e.g. 求 $\int x^2 dx = \frac{1}{3}x^3 + C, C \in \mathbb{R}$. $\int x^{\mu} dx = \frac{1}{\mu+1} x^{\mu+1} + C, C \in \mathbb{R}$.

e.g. 求 $\int \sin x dx = -\cos x + C, C \in \mathbb{R}$. **积分是微分的逆运算.**

2. 不定积分的性质.

$$(1) \int k f(x) dx = k \int f(x) dx = k F(x) + C, C \in \mathbb{R}, k \neq 0.$$

$$(2) \int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx.$$

推论: 若 $f(x) = \sum_{i=1}^n k_i \cdot f_i(x)$, 则 $\int f(x) dx = \sum_{i=1}^n k_i \int f_i(x) dx$.

$$(3) \text{由定义可知: } \frac{d}{dx} [\int f(x) dx] = f(x).$$

$$(4) \text{由定义可知: } \int F'(x) dx = F(x) + C, C \in \mathbb{R} \quad (F'(x) dx = dF(x)).$$

3. 基本积分表.

$$(1) \int k dx = kx + C.$$

$$(2) \int x^{\mu} dx = \frac{1}{\mu+1} x^{\mu+1} + C, \mu \neq -1.$$

$$(3) \int \frac{dx}{x} = \ln|x| + C, x \in \mathbb{R}^*.$$

$$(4) \int \frac{dx}{1+x^2} = \arctan x + C = -\operatorname{arccot} x + C.$$

$$(5) \int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C.$$

$$(6) \int \sin x dx = -\cos x + C.$$

$$(7) \int \cos x dx = \sin x + C.$$

$$(8) \int \sec^2 x dx = \tan x + C.$$

$$(9) \int \csc^2 x dx = -\cot x + C.$$

$$(10) \int \sec x \tan x = \sec x + C.$$

$$(11) \int \csc x \tan x = -\csc x + C.$$

$$(12) \int a^x dx = \frac{a^x}{\ln a} + C.$$

e.g. $\int \tan^2 x dx = \int (\sec^2 x - 1) dx = \tan x - x + C.$

e.g. $\int \frac{x^2+x+1}{x(x^2+1)} dx = \int \frac{x+(x^2+1)}{x(x^2+1)} dx = \int \frac{1}{x^2+1} dx + \int \frac{1}{x} dx = \arctan x + \ln|x| + C.$

e.g. 设曲线经过 $(1, 2)$, 且任一点处切线斜率等于该点横坐标的两倍.

设曲线 $y = f(x) \Rightarrow f'(x) = 2x \therefore df(x) = 2x dx$. 两侧积分: $f(x) = x^2 + C.$

又: $f(1) = 2 \Rightarrow C = 1 \therefore f(x) = x^2 + 1, y = x^2 + 1$. **$(1, 2)$ 称为微分方程的定解条件/初始条件.**

e.g. $\int \frac{x^4}{x^2+1} dx = \int (\frac{x^4-1}{x^2+1} + \frac{1}{x^2+1}) dx = \int (x^2-1) dx + \int \frac{dx}{x^2+1} = \frac{1}{3}x^3 - x + \arctan x + C.$